



SIMATS ENGINEERING



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ASSESSMENT TOOL -2

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CASE STUDY

1. A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an 8% packet loss, 180 ms jitter, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

- 1. Analyze whether the problem is primarily caused by the transport protocol (TCP/UDP) or the application layer.**
- 2. Justify your answer based on the communication requirements of real-time multimedia applications.**
- 3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.**

ANSWER:-

- 1. Is the problem primarily caused by the transport protocol (TCP/UDP) or the application layer?**

The problem is primarily caused by network conditions affecting the transport protocol (UDP), while the application layer plays a secondary role. Cloud-based video conferencing platforms use UDP because it offers low latency, which is essential for real-time communication. However, the reported 8% packet loss, 180 ms jitter, and fluctuating bandwidth result in voice distortion, frozen video frames, and delayed communication. Although the application's buffering strategy can influence playback quality, the main issue is the poor network performance affecting UDP transmission.

- 2. Justify your answer based on the communication requirements of real-time multimedia applications**

Real-time multimedia applications require low latency, low jitter, minimal packet loss, and stable bandwidth for smooth communication. UDP is preferred over TCP because it sends data without waiting for acknowledgments or retransmissions, reducing delay. In contrast, TCP

retransmits lost packets, which increases latency and makes real-time conversations less responsive. Since the network has high packet loss (8%) and high jitter (180 ms), the quality of audio and video is significantly affected, causing distorted voice, frozen video, and communication delays.

3. Recommend suitable protocol-level or application-level improvements

Protocol-Level Improvements:

- Continue using UDP with RTP/RTCP for efficient real-time media delivery.
- Implement Quality of Service (QoS) to prioritize conferencing traffic.
- Enable Forward Error Correction (FEC) to reduce the impact of packet loss.
- Use adaptive bitrate streaming to match video quality with available bandwidth.

Application-Level Improvements:

- Optimize or increase the jitter buffer to handle packet delay variations.
- Implement adaptive jitter buffering based on network conditions.
- Use packet loss concealment (PLC) to improve audio quality.
- Automatically reduce video resolution or frame rate when bandwidth decreases.

Conclusion

The conferencing issues are mainly due to poor network conditions affecting UDP-based communication, while the application's buffering strategy has a smaller impact. Improving network quality through QoS, RTP/RTCP, FEC, and adaptive bitrate, along with optimizing jitter buffering and packet loss concealment, will significantly enhance the quality of real-time video conferencing.

2. A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the average DNS lookup time is approximately 800 ms, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

- 1. Analyze the reasons behind the high DNS resolution time.**
- 2. Propose a suitable DNS architecture using techniques such as Anycast DNS, DNS pre-fetching, and TTL optimization to reduce the lookup time below 50 ms.**
- 3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.**

ANSWER :-

1. Analyze the reasons behind the high DNS resolution time

The 800 ms DNS lookup time is mainly caused by inefficient DNS infrastructure. Possible reasons include:

- DNS servers are located far from users, increasing latency.
- Too many recursive DNS queries due to poor caching.
- High TTL settings preventing efficient cache updates or low TTL causing frequent lookups.
- Overloaded or slow DNS servers.
- Network congestion between users and DNS servers.

These factors delay DNS resolution, resulting in slower webpage loading.

2. Propose a suitable DNS architecture

To reduce DNS lookup time below 50 ms, the company should implement:

- **Anycast DNS:** Deploy multiple DNS servers worldwide using the same IP address so users connect to the nearest server.

- **DNS Pre-fetching:** Allow browsers to resolve domain names before users click links, reducing perceived delay.
- **TTL Optimization:** Set moderate TTL values (e.g., 300–600 seconds) to improve caching while allowing quick updates during server changes.
- **Redundant DNS Servers:** Deploy primary and secondary DNS servers across different regions for failover and high availability.
- **DNS Caching:** Use local recursive resolvers to reduce repeated external lookups.

3. Evaluate the proposed architecture

Advantages:

- Faster DNS resolution (target < 50 ms).
- Improved website performance and user experience.
- High availability through redundant Anycast DNS servers.
- Better scalability to support global users and increasing traffic.
- Reduced server load due to caching and optimized TTL.

Limitations:

- Higher deployment and maintenance costs.
- More complex DNS configuration and monitoring.
- Incorrect TTL settings may lead to outdated cached records or more frequent queries.

Conclusion

Using Anycast DNS, DNS pre-fetching, TTL optimization, and redundant DNS servers can reduce DNS lookup time to below 50 ms while improving performance, scalability, and service availability across multiple continents.

3. A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly 800 ms because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

- 1. Examine the impact of centralized DNS architecture on application performance.**
- 2. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.**
- 3. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance.**

ANSWER :-

1. Examine the impact of centralized DNS architecture on application performance

A centralized DNS architecture directs all DNS requests to a single primary DNS server. During promotional events, millions of users send DNS queries simultaneously, causing the server to become overloaded. As a result, DNS lookup time increases to 800 ms, delaying the loading of cloud-hosted applications.

The major impacts include:

- Increased DNS response time, leading to slow webpage and application loading.
- Reduced user experience because users must wait longer before accessing services.
- Poor scalability, as one server cannot efficiently handle rapidly increasing global traffic.
- Increased latency for users located far from the DNS server because queries must travel longer distances.

2. Design a distributed DNS solution

A distributed DNS architecture should include the following components:

- **Anycast DNS:** Deploy multiple DNS servers across different continents using the same IP address. Users are automatically routed to the nearest available server, reducing lookup time.

- **Primary and Secondary DNS Servers:** Configure multiple authoritative DNS servers in different regions to provide redundancy and automatic failover.
- **DNS Caching:** Use recursive DNS resolvers and local caches so repeated queries are answered without contacting the authoritative server.
- **TTL Optimization:** Configure moderate TTL values (around 300–600 seconds) to improve cache efficiency while allowing timely DNS updates.

This architecture can reduce DNS lookup latency to below 50 ms while maintaining high availability and reliability.

3. Analyze how TTL values, Anycast routing, and DNS caching influence system performance and fault tolerance

TTL (Time to Live):

- Determines how long DNS records remain in cache.
- Higher TTL reduces DNS queries and improves response time.
- Lower TTL allows faster updates during server or IP address changes.
- Moderate TTL provides a balance between performance and flexibility.

Anycast Routing:

- Routes users to the nearest available DNS server using the same IP address.
- Reduces network latency and DNS lookup time.
- Distributes traffic across multiple servers, preventing overload.
- Provides automatic failover if one DNS server fails.

Conclusion

A distributed DNS architecture using Anycast routing, DNS caching, optimized TTL values, redundant DNS servers, load balancing, and health monitoring provides a reliable solution for global cloud services. Compared with a centralized DNS system, it significantly reduces lookup latency, improves application performance, supports millions of users efficiently, and ensures high availability even during promotional events or unexpected server failures.

4. A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the average order acknowledgment latency increases from 1 ms to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

- 1. Analyze three possible TCP-related factors responsible for the latency increase, considering flow control, Nagle's algorithm, and SYN queue limitations.**
- 2. Explain how each factor affects transaction processing in high-frequency systems**
- 3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.**

ANSWER :-

- 1. Analyze three possible TCP-related factors responsible for the latency increase**

The increase in order acknowledgment latency from 1 ms to 40 ms is mainly due to three TCP-related factors:

a) Flow Control:

TCP flow control prevents the sender from overwhelming the receiver by adjusting the receive window size. During peak trading hours, if the server processes orders slowly, the receive window shrinks, reducing the transmission rate and increasing acknowledgment delays.

b) Nagle's Algorithm:

Nagle's algorithm combines multiple small packets into a larger packet before transmission to reduce network overhead. Since stock trading systems send many small order messages that require immediate delivery, this buffering introduces additional delay and increases transaction latency.

c) SYN Queue Limitations:

At market opening, thousands of clients attempt to establish TCP connections simultaneously. If the server's SYN queue is too small, new connection requests are delayed or dropped, leading to retransmissions, longer connection setup times, and increased latency.

2. Explain how each factor affects transaction processing in high-frequency systems

- **Flow Control:** A reduced receive window slows the transfer of trading data, delaying order acknowledgments and reducing transaction throughput.
- **Nagle's Algorithm:** Holding small packets before transmission delays buy and sell orders, which is unacceptable in high-frequency trading where every millisecond matters.
- **SYN Queue Limitations:** A full SYN queue increases connection establishment time, causing delayed trade submissions, retransmissions, and possible connection failures during peak market activity.

These factors reduce the speed and reliability of order processing, leading to delayed trade confirmations and customer dissatisfaction.

3. Recommend appropriate TCP configuration changes

To reduce latency and improve transaction reliability, the following changes are recommended:

- **Disable Nagle's Algorithm (TCP_NODELAY)** so that small order messages are transmitted immediately.
- **Increase the SYN queue (backlog) size** to handle a large number of simultaneous connection requests.
- **Optimize TCP send and receive buffer sizes** to improve flow control and reduce buffering delays.
- **Enable TCP Window Scaling** to support high-speed data transmission efficiently.
- **Adjust TCP Retransmission Timeout (RTO)** to reduce unnecessary retransmissions while maintaining reliability.
- **Use persistent TCP connections (Keep-Alive)** instead of repeatedly opening and closing connections.
- **Enable Selective Acknowledgment (SACK)** to retransmit only lost packets instead of the entire data stream.

Conclusion

The increased latency is primarily caused by TCP flow control, Nagle's algorithm, and SYN queue limitations during peak trading periods. These issues slow order acknowledgments, increase connection delays, and reduce transaction throughput. By disabling Nagle's algorithm, expanding the SYN queue, optimizing TCP buffers, using persistent connections, enabling SACK, and applying QoS, the financial institution can significantly reduce latency, improve transaction reliability, and ensure fast and efficient processing of high-frequency trading orders.